2.1 Rest and Motion

When an object does not change its position with respect to time, then it is at rest. And when an object changes it position with time, then it is in motion.

Now we need to explain the term 'position' properly. In our daily conversation we use the word 'position' in different ways, but in physics the word 'position' has a definite meaning. If you are asked, what is the position of your school and if your answer is 'Jhiltuli', your answer is correct but the position of your school remains unknown. If you reply, your school is 1 kilometer from the gate of your residence; the position of your school is still unknown. Though the position of the gate of your residence is known to us, yet we cannot tell exactly in which direction the school is situated at a distance of 1 kilometer from the gate. But if you say the school is situated 1 kilometer east from the gate of your residence, only then we will know the exact position of your school. That is,to know the position of the school; we have to know both the distance and the direction definitely. Not only this, this distance and direction has to be specified from the position of a reference point. In the case of your school the gate of your residence was the reference point or origin. Instead of your residence gate, the reference point might be a bus stop or a shopping mall. Then definitely both the position and the direction would have different values; but we can specify certainly the position with respect to the new reference point. That is, to specify the position of any object, it has to be mentioned with respect to a reference point. This reference point is not an absolute one; we can conveniently choose any point as reference point or origin.

Now the question is, to specify the position of an object, is it necessary for the reference point or origin to be a static point? Let us think; in front of you a person is sitting still on a chair. If we consider the chair as the reference point or origin, we can firmly say, the position of your friend is not changing.

But if it happens that actually you are sitting in a moving train, what will it then be?

A man outside of the train standing on the station will say, you and your friend are both in motion, nobody is at rest. Then whose statement is true? Your statement or of the man standing in the station? In fact, both are correct! The reason is- if the reference point or the origin moves at a uniform velocity, then we cannot tell firmly whether the reference point is moving at a uniform velocity or actually it is at rest and all other things are moving at uniform velocity in the opposite direction. Therefore, we can say, if an object changes its position with respect to an origin, then the object is in motion with respect to that origin. It is not our headache whether the origin is at rest or moving at uniform velocity. This is not important since every motion is relative.

Not only this, if we want to search out an absolute rest reference point, we will be in trouble. If we consider anything on the surface of the earth as origin, one can object, as earth is not stationary rather it is rotating on its own axis, so everything on its surface is also rotating. Alternatively we can say, the center of the earth is the origin. Then someone else can come up with the objection that the centre of the earth is not stationary, it revolves round the sun. Then we can say more intelligently, the centre of the sun must be the origin! Then another person can confidently say that the sun is also not stationary; it is also revolving around the centre of our galaxy. Surely you are feeling that no one can dare to say, the centre of our galaxy is the origin! Who can say that the galaxy and the universe are stationary? Not only this, if the centre of the galaxy is considered as the origin to describe any position on the surface, do you realize the extent of complexities that may arise?

In fact, we don't need such type of complexities; for our purpose, we can consider any point as the origin, which seems stationary to us. In this case we have to mention all the measurements are done with respect to this origin. In this way scientists have done all the measurements starting from the nucleus of an atom to the satellites launched in space, with no problems what so ever!

2.2 Different Types of Motion

We see various types of motions around us, vibrations, rotations and separations − all these are the examples of different kinds of motions. Probable motions are unlimited, but if we wish, we can talk about some important types of motion separately.

Linear motion

It is an example of the easiest type of motion. If anything moves along a straight line then its motion is a called linear motion. If an object is pushed off on a plane surface then it moves along a straight line. If a ball is allowed to fall from a height, it will fall straight downwards, so it is also a linear motion.

Circular Motion

When a body rotates around a particular point or a line, keeping the distance of the particles of the body unchanged, it is called circular motion. Though the motion of electric fans, the hands of clocks are examples of circular motion, a wonderful example of circular motion is the moon in the sky. The moon is not tied to the earth although it is revolving around the earth, neither is it falling on to the earth's surface.

Translational motion

If an object moves in such a way that all the particles of the object travel the same distance, at the same time, in the same direction then its motion is called translational motion. Sometimes we see many examples of this type in our surroundings. When something moves in a straight path then its example is very common. If we do not consider the circular motion of the wheel of a car then straight advancement of the car is an example of translational motion. Every point of the car will travel the same distance, in equal time, in the same direction.

Figure 2.01: Example of translational motion

There is no obligation that the translational motion should be straight.But the example of translational motion is not easily available in the case of a curved path. Figure 2.01 shows how a plane has to move for every point of the plane travelling equal distances in the same direction. The figure also shows why the example of translational motion on a curved path is so rare.

Periodic Motion

If the motion of a moving object passes repeatedly through a definite point in the same direction in the same manner in a definite interval of time, then this motion is called a periodic motion. The time interval over which this repetition occurs is called the period of the motion. In a periodic motion, a moving particle traverses each point in its path with the same speed after each period.

The vibrational motion of our heart is periodic, since it vibrates in the same direction in the same manner after a definite interval of time. The periodic motion may be circular (motion of the blades of a fan), hyperbolic (orbit of Haley's Comet around the sun) or linear (oscillatory motion of an object hanging from a spring).

Figure 2.02 : Example of simple harmonic motion

Simple Harmonic Motion

Simple harmonic motion is a special type of periodic motion. In case of oscillatory motion the object oscillates on both sides of a definite point. This specific point is called the equilibrium point. At the extreme ends of its path, the particle's velocity is zero. As the particle moves from one end toward the equilibrium point, its velocity gradually increases, reaching a maximum at the equilibrium point. As it continues toward the opposite end, its velocity decreases gradually until it again becomes zero.The particle then reverses direction, moving back in the same manner until it reaches the original end and briefly stops.

There are so many examples of oscillatory motion around us. The motion of an object hanging from a spring is an example of oscillatory motion. The oscillatory baby on a swing (Figure 2.02) or the pendulum of a clock are the examples of oscillatory motion. When we speak, the air molecules carry the sound forward by this type of motion.

So far we have discussed some special types of motion but the causes of these motions have not been mentioned anywhere. The major success of physics is that not only can it find the causes of the different types of motion of objects but it can also explain the motion very clearly.

Can you guess the causes of motion?

2.3 Scalar and Vector quantities

In the world anything that can be measured is called a quantity. Joy and sorrow are not quantities but temperature is a quantity. Because joy and sorrow cannot be assigned a value by measuring them but temperature can be given a value by measurement. The temperature of your body is 37°C or 98.4°F. To express temperature a single number is sufficient, but there are many quantities which cannot be expressed by a single number completely; with its magnitude its direction has to be mentioned or more than one magnitude has to be mentioned so that they altogether can express definitely the magnitude and direction of the quantity. Position is a quantity of this type; to express this only the distance is not sufficient; its direction has to be mentioned too. The quantity which can be expressed only by a single number is called a scalar quantity, on the other hand the quantities for which direction has to be mentioned in addition to its magnitude are called vector quantities.

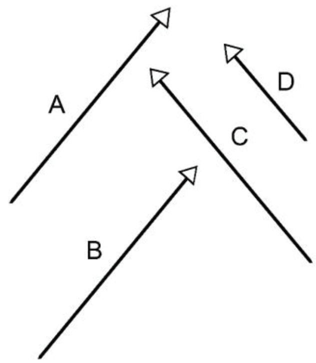

Figure 2.03: Vector A and vector B are equal in all respect although in different position, vector C is different from A and B, because their magnitude is same but direction is different. Vector D is different from C, because direction is same but their magnitude is not same.

Besides temperature examples of scalar are time, length and mass. Because they can be expressed by a single number only. You will find that besides position other examples of vectors are velocity and force. You will be introduced to velocity and force in the next chapter. Because to express these quantities, direction is to be mentioned along with its magnitude.

To differentiate vector quantities from scalar quantities they are written in bold font (e.g. **x**, **y** or **A**, **B**). In books or in computer printing it is easy to write anything in bold form. But when anyone writes on a paper then to express anything as vector, a small arrow is used over it ($\vec{x}, \vec{y}$ or $\vec{A}, \vec{B}$).

In this book the physics you will be taught will not involve application of vectors. At best you will be reminded which one is scalar and which one is a vector.

2.4 Distance and Displacement

We are quite familiar with the word 'distance' but we do not use the word 'displacement' in our daily life. We want to understand the relation between the two words- distance and displacement with the help of an example. A curved path is shown in figure 2.04. The distance travelled with respect to the point A is denoted by the number 1, 2, 3 in kilometer.

Let us consider you are at point A (i.e. your position is point A). Now you have reached point B 4 km away by riding a bicycle along the curved path. We can say the distance between point A and point B is 4 km. Distance is a scalar quantity, so to express the distance between the points A and B we need not mention any direction. We determined the 'distance' of the point B along this path with respect to the point A. Now if we desire, we can determine the 'displacement' of point B with respect to point A. By displacement the position of point B with respect to point A is meant. This can be expressed as $\overrightarrow{AB}$ as well.

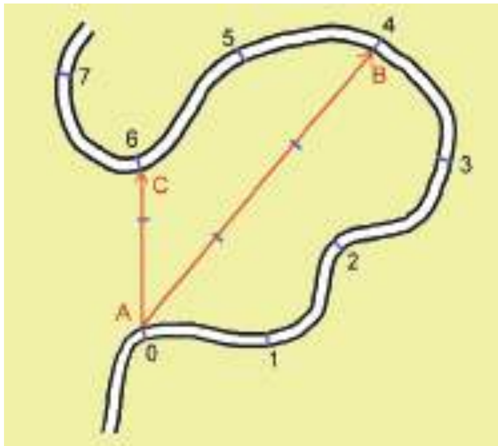

Figure 2.04 : Curved path trakelled by a cycle from point A.

In the figure the displacement is shown by a straight line from point A to point B with an arrowhead. In this figure the magnitude of the displacement is 3 km and the direction of the arrow is in the direction of the displacement. Therefore displacement is a vector quantity; it has both magnitude and direction.

If going two kilometers more by a cycle you cross a six kilometer path total and reach point C, then your displacement will be the straight line $\overrightarrow{AC}$ with an arrow, whose magnitude is 1.5 kilometer and the direction of the arrow is along your displacement. Though you covered more distance along the curved path but your displacement is still small. Therefore it is not true that the more distance you cover, the more will be the displacement.The difference between the initial and the final position is the displacement.

Starting from the point A, the distance towards point B is 4 km along the curved path. Similarly the distance between point B and point A is 4 km i.e. both are equal. But notice that the displacement from A to B and the displacement from B to A are not equal. One is negative of the other. In vector form we can write,

$$\overrightarrow{AB} = -\overrightarrow{BA}$$

Distance or displacement, both are the dimension of length.

[Displacement] = L (vector)

[Distance]= L (scalar)

2.5 Speed and Velocity

Roughly we know what is meant by velocity. The measurement of how fast a body is moving is called the velocity. In physics, velocity has a definite meaning and in addition to velocity we use another quantity named speed. If we have understood the two terms distance and displacement properly, then we will be able to understand the two terms speed and velocity very easily.

Speed is the rate of change of distance with respect to time. Therefore, if you have covered 100 m distance in 20 second, your speed will be;

$$v = \frac{100\ \text{m}}{20\ \text{s}} = 5\ \text{m/s}$$

The dimension of speed $[v] = LT^{-1}$

Velocity is the rate of change of displacement with respect to time. Therefore, if the change of your position along a certain direction is 50 m in 20 second, then magnitude of your velocity will be;

$$v = \frac{50\ \text{m}}{20\ \text{s}} = 2.5\ \text{m/s}$$

And that certain direction is the direction of velocity.
As velocity is a vector quantity, we will have to fix its direction.

The dimension of velocity $[v] = LT^{-1}$

Here, one thing has to be noticed. If we consider only linear motion, then there is no difference between velocity and speed, the magnitude of velocity is the speed. To understand the internal relationship between speed and velocity, some examples are discussed below.

To explain distance and displacement we considered a curved path in figure 2.04 and showed different positions in that path. To have an understanding about of speed and velocity, we can consider the same example. But now we have to tell how much time you have taken for going from one position to another. Suppose you have taken 20 minutes for coming from position A to B by bicycle. Then your average speed will be:

Average speed = Total distance travelled/ Total time

Therefore,

$$v = \frac{4\ \text{km}}{20\ \text{minutes}} = \frac{4 \times 1000\ \text{m}}{20 \times 60\ \text{s}} = 3.33\ \text{m/s}$$

Notice that, we have used the word average speed instead of the word speed. Because, while you were riding the bicycle, sometimes you rode fast and sometimes you rode slow. So we cannot talk about the 'instantaneous' speed, rather we can talk about the average speed in the time interval.

Let us try to determine velocity now. Like speed, we cannot calculate instantaneous velocity. In this time interval, you rode the bicycle at different velocities. The velocity has changed due to the motion being fast or slow. Again the velocity has changed due to the change of direction. Considering these changes, the magnitude of the average velocity will be:

Average velocity = displacement/time

Magnitude of average velocity = Magnitude of displacement/time

Therefore,

$$v = \frac{3\ \text{km}}{20\ \text{minutes}} = \frac{3 \times 1000\ \text{m}}{20 \times 60\ \text{s}} = 2.5\ \text{m/s}$$

In this example you have noticed that the value of the average velocity is less than that of the average speed. If the path was straight rather than curved, then the magnitude of the average velocity would be equal to the magnitude of the average speed.

In our example, if you rode your bicycle always at the same speed t hen we can say your speed is uniform speed. When anything moves with uniform speed then its instantaneous speed and average speed will be the same.

Notice that, since the path is curved, if you go through this path your direction is changing continuously. Therefore you can move with uniform speed through this path but you cannot go with uniform velocity. If anything moves in a linear motion along a straight line, only then uniform velocity or constant velocity is possible.

Example

Question: Let us consider another example to understand the relation between velocity and speed more clearly. Tie a small piece of stone with a string and rotate it around head (figure 2.05). Does the stone move with uniform velocity or with uniform speed?

Answer: If you think for a while you will be able to understand that the speed of the stone is not changing but the velocity is changing at every instant! Because at every instant the direction of motion of the stone is changing. If the stone moves along a straight line, then the direction of its motion will not change. Since it is revolving, its direction of motion is changing continuously. So it is an example of uniform speed- not of uniform velocity. If it is uniform velocity, then it must be uniform speed. Whereas for uniform speed it is not guaranteed that the velocity will be uniform.

Figure 2.05 : In case of rotation of a stone tied with a string, velocity might change though speed is same.

Question: If the stone is suddenly released, will it move with uniform velocity or with uniform speed?

Answer: If the stone is released suddenly, it will move straight with uniform velocity and uniform speed. It will continue to move with uniform velocity and uniform speed if there is no air friction or gravitational force etc.

2.6 Acceleration

When an object moves with uniform velocity it has no acceleration. If there is a change of velocity, then there is an acceleration. More clearly we can say, the rate of change of velocity with time is acceleration.

Since velocity has both magnitude and direction, the change of velocity can occur in two ways. In our previous example, when you rode your bicycle along the curved path, the change of velocity occured at every turn and you accelerated. Although you travelled the whole path with uniform speed, acceleration took place due to change of direction only. If you tied a stone with a string, as in the previous example, and rotated it over your head with uniform speed then the rotating stone would change its direction of motion continuously. That is, change of velocity would occur and acceleration would takes place.

If your motion is linear, then there is no scope of changing the direction. Then acceleration may occur only for the change of magnitude of the velocities (speed). When the magnitude of the velocity increases, then we can say positive acceleration is taking place along the direction of the velocity. If the velocity decreases, then we can say negative acceleration or deceleration is taking place. Now we can determine the acceleration of an object moving along a straight line.

Example

Question: In figure 2.06 the change of velocity of an object with time is shown. Identify where there is acceleration and where it is absent.

Answer: There is acceleration at A; at B there is no acceleration. At C, there is acceleration; at D there is negative acceleration or deceleration.

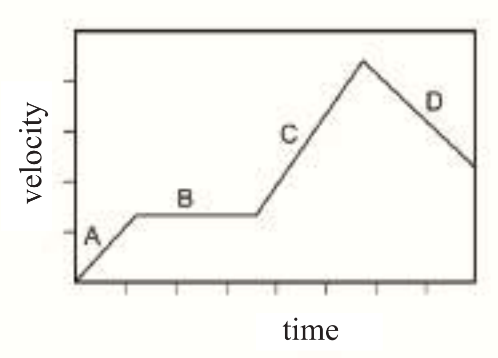

Figure 2.06: Change of velocity of an object with time

In this chapter we will discuss only the linear motion in which only the change in magnitude of velocity causes acceleration to take place.

Acceleration is the rate of change of velocity with time. If the acceleration is uniform i.e. it will not change with time, then we can write;

$$\textbf{acceleration} = \frac{\textbf{(final velocity-initial velocity)}}{\textbf{time}}$$

If the velocity of an object at a certain time is u and after time t the velocity becomes v, then the acceleration a will be;

$$a = \frac{v - u}{t}$$

The dimension of acceleration: [a] = LT^{-2}
Unit of acceleration: ms^{-2}

When the acceleration a is known and if the initial velocity is u then after time t it is very easy to find the final velocity v (Figure 2.07). The final velocity is,

$$\boxed{v = u + at}$$

If the object starts its motion with the same acceleration from rest, then

$$v = at$$

We have alrady said that all the discussions done so for are true for uniform acceleration If the object does not move with uniform acceleration, then it is not so easy to find the acceleration from the initial and final velocities.

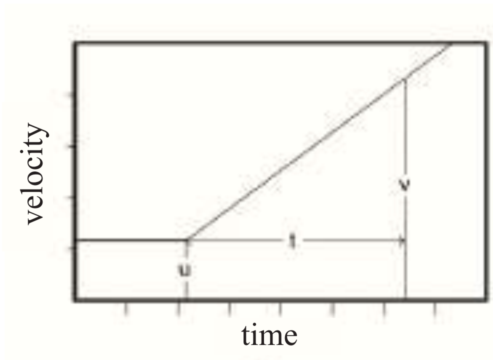

Figure 2.07 : Increasing velocity of an object which is moving with uniform acceleration, started from rest.

The examples of motion we see around us, such as the motion of a car, train or a bicycle etc. have their accelerations as non-uniform almost all the time. For example, if a car starts its motion from rest and increases its velocity gradually, then starting from zero its acceleration reaches a particular value. When the car attains its peak velocity, then its velocity does not increase any further and the acceleration becomes zero again. If the car reduces its velocity then retardation starts. If the car ceases its motion, then both its velocity and acceleration become zero. You may think that the examples of uniform acceleration are rare. There is an amazing example of uniform acceleration which we see in our surroundings. That is the acceleration due to gravity. Near the earth's surface, the value of this acceleration is 9.8 ms^2. If we release an object from rest from above the earth, then its velocity increases according to the equation v = gt.

2.7 Equations of Motion

Since we discuss only linear motion, the quantities we have talked about so far are:

u: initial velocity

a: acceleration

t: elapsed time

v: velocity after elapsed time

s: distance covered in elapsed timed

Except time (t) the magnitude of these quantities may be positive or negative along a strait line We will not consider them as vectors and will talk about their magnitude only.

Discussions about almost al the relationship among these quantities, are done already, only one is left, which is the relationship for the distances. If an object has no acceleration, then there is no change in its velocity. Then the initial velocity and the final velocity will be equal i.e.(u = v). Therefore the distance covered will be,

$$s = v t$$

If there is uniform acceleration, the final velocity is:

$$v = u + at$$

This equation shows that the velocity is changing with time. If the entire duration is divided into many tiny equal intervals, the distance covered in each small interval will not be the same. This is because the particle's velocity changes constantly due to acceleration. However, to calculate the total distance traveled, these varying distances must be summed. Calculus, a specialized branch of mathematics, is used to perform such calculations. However, if the particle's acceleration is uniform, it is possible to determine the total distance traveled without using calculus

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Since the velocity is changing every moment hence we cannot write the equation $s = vt$ but if we consider an average velocity V then we can write

$$s = Vt$$

In the case of a uniformly accelerated particle, the average velocity over a specific time interval is equal to the velocity at the midpoint of that interval.

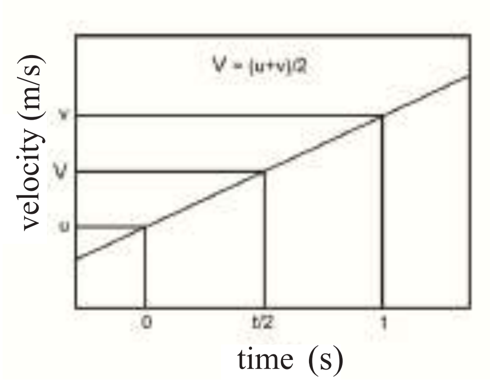

Figure 2.08: Average velocity is the mid time velocity of the initial and final velocity.

$$V = \frac{u+v}{2} = \frac{u+(u+at)}{2}$$

$$V = u + \frac{1}{2}at$$

Therefore the travelled distance is,

$$s = Vt$$

$$s = \left(u + \frac{1}{2}at\right)t$$

$$\boxed{s = ut + \frac{1}{2}at^2}$$

In all the equations of motion we have already deduced the time t is present. We can deduce another equation in which t is absent.

Such as:

$$v = u + at$$

$$v^2 = (u+at)^2 = u^2 + 2uat + a^2t^2 = u^2 + 2a\left(ut + \frac{1}{2}at^2\right)$$

$$\boxed{v^2 = u^2 + 2as}$$

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Although this equation looks like another general equation, there is some amazing physics hidden in it, which we will reveal in the fourth chapter.

Example

Question: The velocity of a car is increased to 60 km/hour within 1 minute of starting from rest. What is its acceleration?

Answer: From now we will use the unit of time in second (s) instead of minute or hour and for distance meter (m) will be used instead of mile or kilometer.

The final velocity of the car is

$$v = 60\,\frac{\text{km}}{\text{hour}} = \frac{60 \times 1000\ \text{m}}{60 \times 60\ \text{s}} = 16.67\ \text{m/s}$$

So the problem is like this, a car attains a velocity of 16.67 m/s in 60 s starting from rest, what is its acceleration?

$$v = at$$

$$a = \frac{v}{t} = \frac{16.67\ \text{m/s}}{60\ \text{s}} = 0.278\ \text{m/s}^2$$

Question: A car is moving with a velocity of 60 mile/hour, suddenly its engine stops. It takes 5 minutes to come at rest. What is the deceleration of the car?

Answer: When there is acceleration, velocity increases; whereas velocity decrease when there is negative acceleration or deceleration.

Again we will use *s* for time and *m* for distance.

$$1\ \text{mile} = 1.6\ \text{km} = 1600\ \text{m}$$

Initial velocity of car,

$$u = 60\,\frac{\text{miles}}{\text{hour}} = \frac{60 \times 1.6 \times 1000\ \text{m}}{60 \times 60\ \text{s}} = 26.8\ \text{m/s}$$

Final velocity of car, $v = 0\ \text{m/s}$

Acceleration,

$$a = \frac{v-u}{t} = \frac{0 - 26.8\ \mathrm{m/s}}{300\ \mathrm{s}} = -0.089\ \mathrm{m/s^2}$$

Thus the acceleration of the car $-0.089\ \mathrm{m/s^2}$ or deceleration $0.089\ \mathrm{m/s^2}$

Question: The velocity of a bullet is 1.5 km/s. It has penetrated 10 cm of a wall. What is the deceleration of the bullet?
Answer: The only way to solve this problem is to use the formula

$$v^2 = u^2 - 2as$$

Final velocity, $v = 0$

$$0 = (1.5 \times 1000\ \mathrm{m/s})^2 - 2a\left(\frac{10}{100}\mathrm{m}\right)$$

$$a = \frac{(1.5 \times 1000)^2}{0.2}\ \mathrm{m/s^2} = 11{,}250{,}000\ \mathrm{m/s^2}$$

Deceleration: $11{,}250{,}000\ \mathrm{m/s^2}$ (or acceleration: $-11{,}250{,}000\ \mathrm{m/s^2}$)

2.8 Laws of Falling Bodies

We have already mentioned that an amazing example of uniform acceleration is acceleration due to gravity *g*. Due to its effect an object falls downward when it is released from above the earth surface. Observing these types of falling bodies Galileo invented three laws. The laws are used in case of bodies falling freely. The laws are as follows:

First law: All bodies falling from rest and from the same height without any resistance traverse equal distance in the same time.

Second law: The velocity (v), acquired by a freely falling body from rest in a given time (t) is directly proportional to that time. i.e. $v \propto t$

Third law: The distance (h) traversed by a freely falling body from rest in a given time (t) is directly proportional to the square of the given time. i.e. $h \propto t^2$

We have mentioned earlier that the acceleration due to gravity is an example of uniform acceleration. The equations we have deduced about motion, can be used to deduce the equations of motion of the falling bodies. In case of motion s was used to indicate the travelled distance, For a falling body we will use h to indicate height. For acceleration we will use g instead of a. These two will be the only difference!

$$v = u + gt$$

$$h = ut + \frac{1}{2}gt^2$$

$$v^2 = u^2 + 2gh$$

The three laws of falling bodies of Galileo are nothing but these equations of motion.

The first law states that all objects released from the same height will reach the ground at the same time i.e. it does not depend on the mass of the objects. This does not go with the experience of our daily lives. If a piece of paper and a piece of stone are released from the same height, at the same time, then the stone reaches the ground first and the paper reaches the ground later. This happens due to the resistance of air. If the experiment is done in a vacuum tube then both the paper and stone will reach the ground at the same time. Galileo's first law can be understood from the equation of falling bodies. This is because there is no mass of object in the equations of velocity and traversed height. That is the acceleration due to gravity acts equally on both heavy and light objects. So the freely falling body traverses equal distances in the same time.

The second law of Galileo is the law of increase of velocity due to g. If the initial velocity u is zero then velocity v is proportional to g. Galileo's third law is nothing but the equation of h. In this formula if we consider $u = 0$ then we see that traversed distance h is proportional to t^2.

Example

Queston: A good pace bowler of cricket can throw a ball with a velocity of 150km/hour. If he throws the ball vertically upwards, how high will it go?

Answer:

$$150 \text{ km/hour} = \frac{150 \times 1000 \text{ m}}{60 \times 60 \text{ s}} = 41.67 \text{ m/s}$$

Acceleration due to gravity will act as retardation when the ball is thrown vertically upwards. The ball eventually comes to a stop. If the height is expressed by h then,

$$v^2 = u^2 - 2gh$$

$$v = 0, \qquad u = 41.67 \text{ m/s}, \qquad g = 9.8 \text{ m}/s^2$$

Therefore, $$h = \frac{u^2}{2g} = \frac{(41.67)^2}{2 \times 9.8} \text{ m} = 88.59 \text{ m}$$

[The ball will approximately reach the roof of a 30 storied building]

Question: When a space ship revolves round the earth, its speed is very high and is nearly 10 km/s. If a cannon ball is fired at this velocity straight upward, how high will it go?

Answer: Let us try as we did with the cricket ball, only the initial velocity will be 10,000 m/s instead of 41.67 m/s.

Therefore, $$h = \frac{(10{,}000)^2}{2 \times 9.8} \text{ m} = 5{,}102{,}000 \text{ m} = 5{,}102 \text{ km}$$

Though it seems that there is no mistake anywhere but actually the answer is not correct. This is because we have considered the value of acceleration due to gravity is 9.8 m/s^2, this is true for the distances near the surface of earth. But if we go far from the earth the value of g will decrease. When we deduced the equation

$$v^2 = u^2 - 2gh$$

then we considered that the value of g is not changing. It is not true for this problem. So what we have learned till now, that knowledge cannot be used to solve this problem. If we cannot solve it, there will be no loss; because if anything is thrown upwards with this high velocity it will burn due to air friction!

Do yourself

Determination of velocity and acceleration at any time from a distance -time graph.

(Motion and Graph)

We have deduced equations of motion in the previous sections. We have analyzed the relations between distance travelled, velocity and acceleration. In this section we will analyze the same quantities but only by graph. We can have a kind of real feelings about different quantities of motion, if we analyze them graphically.

Table 2:01

Time (s)	Distance (m)
0	0
1	1
2	4
3	9
4	16
5	25

Time (s)	Distance (m)
0	0
2	6
4	24
6	54
8	96
10	150

We need to mention something here. Whenever we have discussed distances travelled, velocity or acceleration we always considered a standard situation. We considered that when an object moves, there is no friction and the object does not lose its energy by any other means. That does not happen in real life. That is why it is not so easy to collect real data for distances travelled, velocity or acceleration. For performing a real experiment an air track is used in the laboratory, where an object is kept floating in an air layer so that no friction is present. To measure the change of position of an object with respect to time, electric sparks or electronic signals are used. We will not get such type of data in our daily lives easily. For now we will consider that we have collected some data in this standard situation to use in our graph. Two sets of data have been shown in the table

2.01 which represents the change of position of an object with respect to time. We will solve the first set for you and you will solve the second one by yourself.

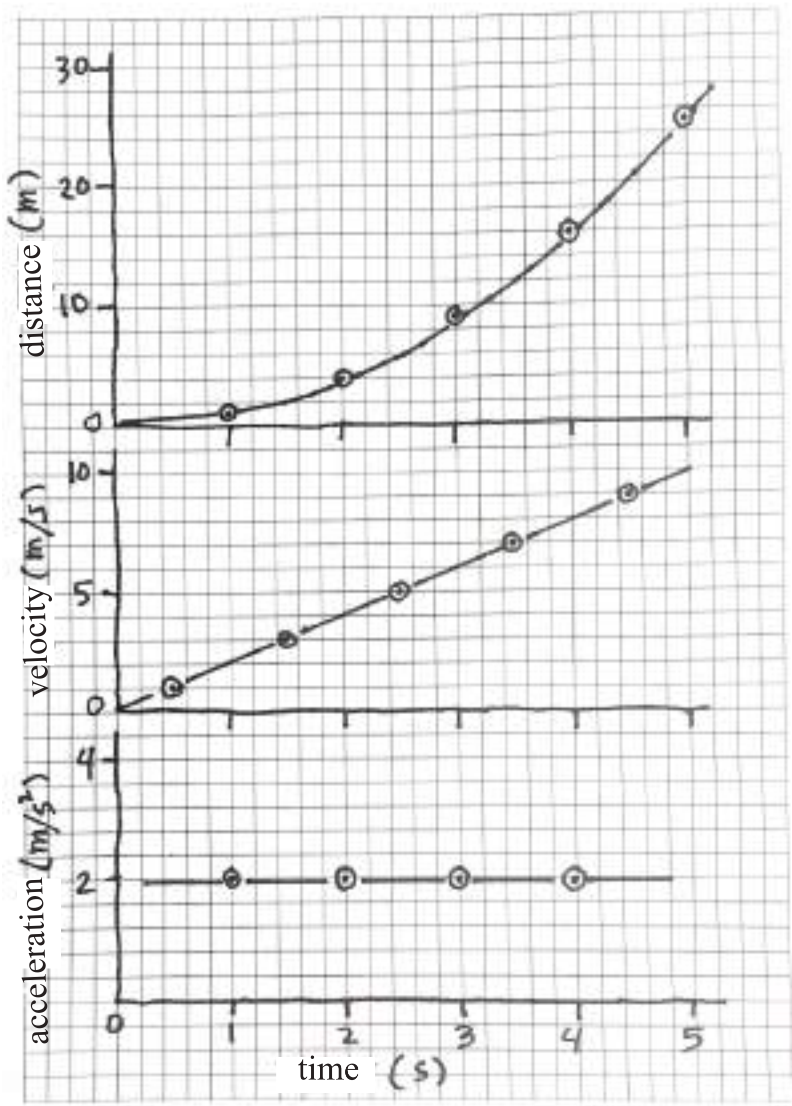

Figure 2.09: Finding velocity - time from distance - time and acceleration - time from velocity-time graphs are drawn.

The distance – time graph of the first data set of the table is shown in figure 2.09. We have taken data in integer form. From the graph we will be able to find the distance over any time from 0 to 5s. For example the distance of the object in 2.5 second is approximately 6.25m. If we have a velocity versus time graph, from there we can find the velocity easily. Velocity of an object is the rate of change of its position. So we can see from the graph the object has travelled a distance of 0 m to 1 m in 0 to 1 second. Therefore the average velocity at this time,

$$v = \frac{(1-0)\ \text{m}}{1\ \text{s}} = 1\ \text{m/s}$$

We can use this average velocity for the mid-time value of 0 to 1s. Similarly the average velocity within 1 to 2s is,

$$v = \frac{(4-1)\ \text{m}}{(2-1)\ \text{s}} = 3\ \text{m/s}$$

We can use this velocity as data for the time 1.5 s within 1 to 2s. Similarly we see the average velocity between 2 s and 3 s is 5 m/s, between 3 and 4 s is 7 m/s and between 4 s and 5 s is 9 m/s. We see these data points are on a straight line on a graph paper and we can connect the points by drawing a straight line. Although we have put the data for time 0.5, 1.5, 2.5, 3.5 and 4.5 s, but we can find the velocity at any time after drawing a straight line through these points. For example at time 3 s the velocity is 6 m/s.

We will be able to find the acceleration in the same way after drawing the velocity -time graph of figure 2.09. Acceleration is the rate of change of velocity. Since the velocity- time graph is a straight line, so in this case we will get the same value of the acceleration at any point on the graph. For example the rate of change of velocity in between the time 2 s to 3 s is,

$$a = \frac{(6-4)\ \text{m/s}}{(3-2)\ \text{s}} = 2\ \text{m/s}^2$$

For any other time if we calculate the acceleration, the same value will be found. It is shown in the acceleration- time graph in figure 2.09.

Therefore, you have seen that we have found velocity or acceleration at any time starting from the distance- time graph. The more accurately we draw this graph, the more precisely we can calculate those quantities. Now, use the second set to calculate velocity and acceleration from graph.

Investigation 2.01

Determination of the average speed of an object rolling over a stope.

Objective: To determine the speed for the same distance travelled on different slopes and then to find relation with slope with the help of graph.

Apparatus:

1. A plane plank or bench or table.
2. A ruler or meter scale.
3. A marble or a cylindrical pen or a pencil which can roll.

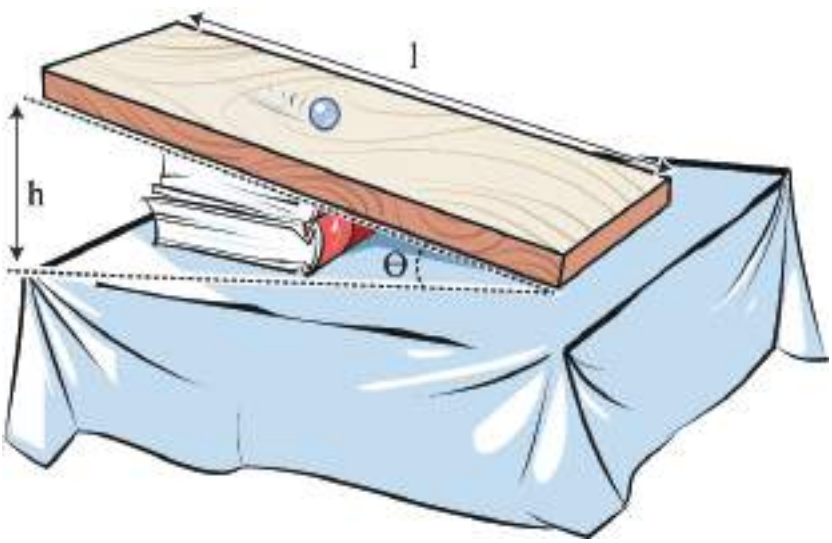

Figure 2.10: A marble is allowed to roll on an inclined plane.

Procedure:

1. Taking a plank or a bench or a table, measure its length (L) by a ruler or meter scale. This distance will be the distance travelled by the object.

2. By placing a book under one end of the plane plank or bench or table let us make it inclined. Measure the height (h) of that end Dividing height by length, find the slope ($\sin\theta = h/L$).

3. Put a marble or a pencil or a pen on the inclined plane and make sure that it can roll.

4. You have to measure the time taken by the marble to travel the length of the inclined plane. A stop watch is needed to measure the time accurately. But it is unlikely for you to have a stop watch. (Now a days stop watch option is available in many mobile phones.) Use the method given below. If you have a normal watch instead of a stop watch, you will be at a disadvantage. Because normal watch cannot measure less than a second. We need to measure more accurately. If we have no stop watch, we can try to measure the time by other means. How many numbers (one, two, three,.....) can you count in fifteen seconds at normal speed?
Let us assume that you can count up to forty-five in fifteen seconds. In that case, you can utter every number in approximately 15/45 = 1/3 second. Let the marble or pencil or pen move on the inclined plane and start counting one, two, three and so on find up to which number you could count for the the marble, pencil or pen to reach the lower end of the inclined plane. Calculate the actual time by multiplying with the correct multiple.

5. Repeat the experiment several times and calculate the average time.

6. Calculate the speed, dividing the length of inclined plane by the average time. This is average speed.

7. Increase the slope of the inclined plane by placing another book. Measure the new height after placing the second book. Calculate the slope for this height.

8. Let the marble, pencil or pen roll on the inclined plane again. Measure the time by counting the numbers and calculate the speed again. Increase the slope gradually and calculate the average speed every time.
9. Draw a graph by plotting sinθ along the X-axis and average speed along Y-axis. From the graph calculate the speed for any slope.

No. of Readings	Distance L cm	Height h cm	$\sin\theta = h/L$	Time t s	Speed $= \frac{\text{Distance}}{\text{Time}}$ m/s	Average speed m/s
1 2 3						
1 2 3						
1 2 3						

Discussion: Describe the relation between slope and speed.
Discuss what necessary measures should be taken to perform the experiment with more accuracy.

Investigation 2.02

Playing with different kinds of motion.

Objective: To find the differences among different kinds of motion through games.

Requirement : Small open space.

Procedure:

1. You have to select definite activities to explain different kinds of motion.

   Linear motion: You have to run straight. If you face any obstructions you have to turn back and go straight again.

   Rotational motion: Spreading your hands on opposited sides start roatating.

Translatory motion: Looking in a particular direction move back and forth and left and right.

Periodic motion: You have to run along a circular path and repeat several times.

Oscillatory motion: Raise your hands and oscillate them left and right.

2. Those who are interested in playing this game will stand scattered around the room.
3. An instructor will loudly say the words linear, rotational, translatory, periodic or oscillatory motion.
4. Everyone has to perform the activities of the game as instructed by the instructor. Those who will fail to follow the instructions will be knocked out.
5. The instructor of the game will utter different motions at different times and the boys and girls have to act out that motion.

The person who will be able to demonstrate all kinds of motion perfectly, he or she will be declared the winner.

Keeping the fundamental characteristics of the motions unchanged, the activities of different motions can be changed as necessary. For example, the instructions may be given to demonstrate two different motions at the same time. In case of linear and oscillatory motion the participant has to run straight by oscillating his or her hands left and right.

Discussion: Write down some additional activities that can demonstrate different motions.

Investigation 2.03

Determining the speed of moving vehicles.

Objective: To determine the speed of different kinds of vehicles by measuring the distances travelled at different times.

Apparatus: Ruler.

Working Procedure:

1. To determine the speed of a vehicle, at first you have to measure the distance between two stationary objects beside a road (e.g. light post, tree, shop etc). Measuring this distance very accurately may be difficult. That's why we will use a simple way to do it. At first you have to measure the length of your step by a ruler. (Measure the distance of ten steps and divide it by ten to make it accurate).

2. Now walk from one stationary object to another object on road sides. Count the number of steps you need to travel the distance and multiply it by the distance of your single step to calculate the distance. It will be better if the distance is approximately hundred meters.

3. Now try to measure the speed of a bicycle, rickshaw, tempo or a pedestrian from a safe distance beside the road. Since the distance is known, you can measure the required time, and then speed can be calculated.

4. To measure the time accurately a stop watch is required, or an ordinary watch will meet up the purpose instead of a stop watch. If you have none, you can use an easy method to measure the time. It takes almost one second to utter the words "one thousand one", "one thousand two", "one thousand three" etc. Counting like this we can measure it.

5. When you will see a bicycle, rickshaw, tempo or a pedestrian just passing the first stationary object, you start to count time with a watch, or start counting "one thousand one", "one thousand two" and so on. Again when you will see this vehicle just passing the second stationary object, you see the time on the watch or stop counting the numbers. Find the required time from the watch or up to which number of words you have counted, This is the time to travel the distance.

Calculate the speed dividing the distance by the time.

Discussion: Compare your procedure of measuring time with the time of a watch and see how accurate it is. Guess the percentage of error of your calculated speed considering the degree of accuracy of your step measuring procedure.

Vehicles	No. of steps	Distanced travelled L (m)	Time t (s)	Average speed = L/t (m/s)